# Multimodal Contrastive Alignment of Histopathology and Bulk Transcriptomics in Breast Cancer

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# Multimodal Contrastive Alignment of Histopathology and Bulk Transcriptomics in Breast Cancer

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***Abstract*— Routine H&E histopathology slides are widely available for breast cancer patients, whereas transcriptomic profiling remains more costly and less routinely accessible. This gap limits the scalability of molecularly informed precision oncology. We propose a diagnosis-label-free contrastive learning framework that aligns tumor-region H&E histopathology features with matched bulk RNA-seq transcriptomic profiles in a shared latent space. Using paired histopathology patches and transcriptomic profiles from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort, the model learns patient-level image–transcriptome correspondences through a dual-encoder InfoNCE objective. We evaluate the learned space through cross-modal retrieval of gene expression, MSigDB Hallmark pathway activity, and PAM50 molecular subtype information. The model retrieves modest but biologically meaningful transcriptomic and pathway-level signals from histopathology alone, including concordance for selected cancer-relevant pathways. These results provide a reproducible proof of concept for image-based transcriptomic profiling from routine histopathology, while further benchmarking, external validation, and clinical utility assessment are required before clinical use.**

## I. Introduction

Precision oncology requires understanding the relationship between molecular tumor states and tissue morphology [1]. In breast cancer, transcriptomic profiling can provide detailed information about tumor biology, molecular subtype, and pathway activity. However, transcriptomic sequencing remains more costly and less routinely available than standard H&E histopathology, which is generated for most patients as part of routine diagnostic workflows. This creates an important computational opportunity: if molecularly informative signals can be inferred from routine histopathology, archived tissue slides could support scalable molecular profiling, patient retrieval, and biomarker discovery.

Traditional computational approaches often treat histopathology and transcriptomics as isolated modalities [2]. For instance, standard workflows typically analyze Whole Slide Images (WSIs) independently to predict clinical outcomes or cellular composition, while parallel pipelines analyze gene expression profiles in isolation to determine molecular characteristics [3, 4]. When multimodal data are used, conventional methods often rely on late-fusion strategies that concatenate extracted visual features with tabular molecular features for downstream prediction tasks [5, 6]. Such approaches may not explicitly model the correspondence between tissue morphology and transcriptomic state.

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Several prior studies have demonstrated considerable potential for inferring gene expression from histopathology images, at both the tissue level and in spatially resolved contexts. Earlier methods predominantly framed this task as a supervised learning problem, using histopathology images as input features and gene expression values as prediction targets [7–9]. More recent methods have introduced multimodal and contrastive frameworks, particularly for spatially resolved transcriptomic prediction [10, 11]. In contrast, our study focuses on a lightweight patient-level contrastive retrieval framework that aligns tumor-region histopathology features with matched bulk RNA-seq profiles and evaluates the resulting shared space for gene-expression, pathway-level, and molecular subtype retrieval.

Here, we propose HistoGenX, a diagnosis-label-free dual-encoder contrastive learning framework for aligning breast cancer histopathology features with bulk transcriptomic profiles in a shared latent space. The model learns patient-level image–transcriptome correspondences from paired H&E-derived tumor-region features and matched RNA-seq profiles from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort, without using PAM50 subtype labels during contrastive representation learning [12].

We evaluate whether the learned embedding space supports three downstream tasks: transcriptomic profile retrieval, MSigDB Hallmark pathway activity retrieval, and PAM50 molecular subtype retrieval. Together, these analyses test whether routine histopathology-derived features can recover biologically meaningful molecular information from bulk transcriptomic profiles.

## II. Methods

### A. *Data collection and pre-processing*

We utilized paired transcriptomic and histopathology data from the TCGA Pan-Cancer analysis project [12], focusing on the TCGA-BRCA cohort. Data were paired at the patient level and filtered for primary tumor samples. Histopathology and transcriptomic records were matched using the TCGA patient barcode, defined by the first 12 characters of the TCGA sample identifier. Using unique TCGA patient IDs to align modalities, we used the predefined internal split from the TCGA-UT dataset on Hugging Face [13]. This split comprises 513 patients for training, 110 for validation, and 111 for testing, corresponding to 16,580, 3,550, and 3,560 image patches, respectively. All splits were patient-disjoint, with no TCGA patient ID appearing in more than one split, to prevent data leakage across training, validation, and test sets.

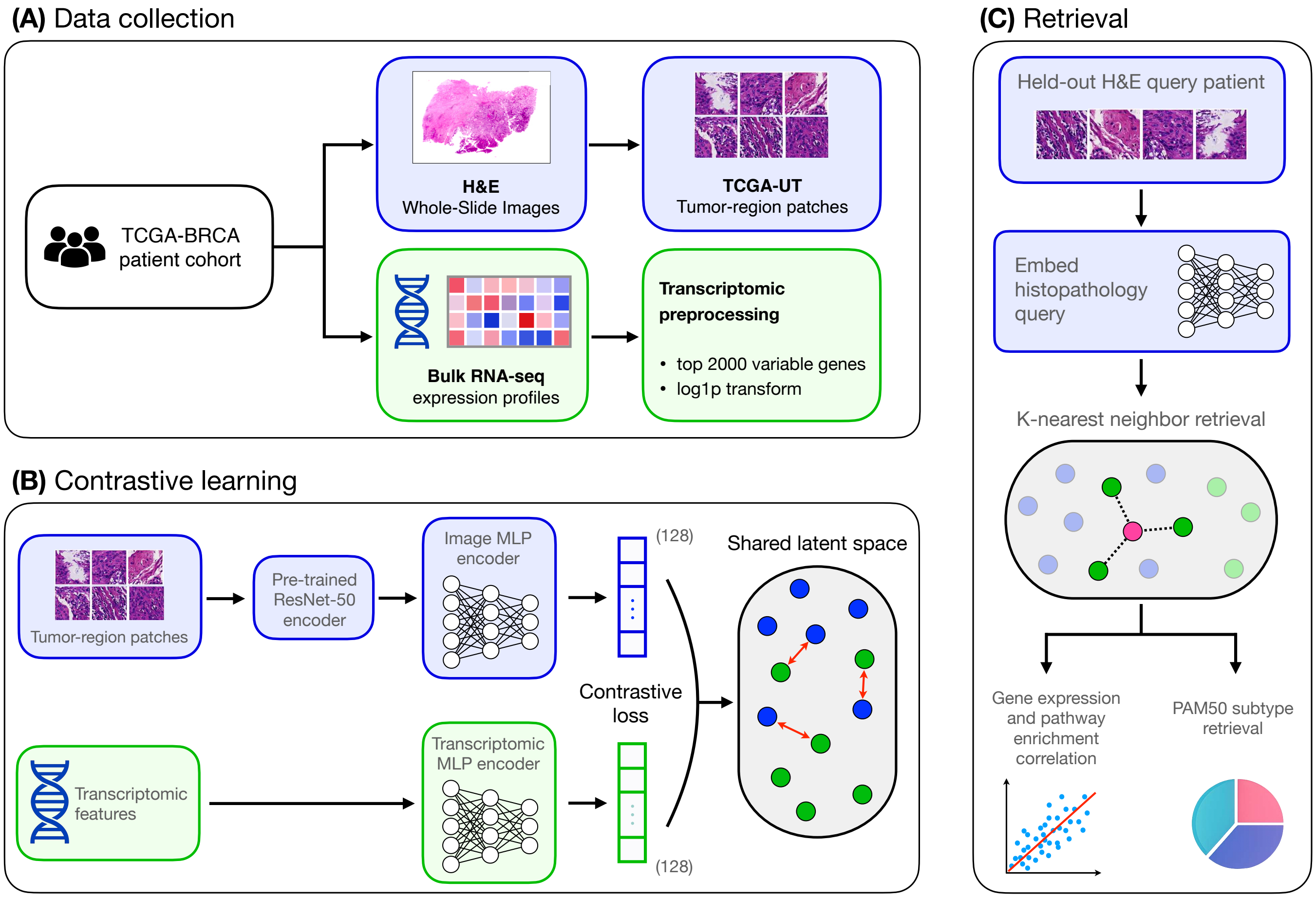


**Figure 1:** Overview of the HistoGenX framework. (A) Paired TCGA-BRCA histopathology and bulk RNA-seq data are prepared using TCGA-UT tumor-region patches and preprocessed transcriptomic profiles. (B) A dual-encoder contrastive model maps patient-level histopathology and transcriptomic features into a shared 128-dimensional latent space, aligning matched image–RNA pairs while separating mismatched pairs. (C) For held-out patients, histopathology embeddings are used for cross-modal nearest-neighbor retrieval of transcriptomic profiles, followed by gene-expression, pathway-level, and PAM50 subtype evaluation.

*a) **Histopathology modality processing**:* We utilized the preprocessed *TCGA-UT* dataset (*Histology images from uniform tumor regions in TCGA Whole Slide Images*) [13]. This dataset contains Whole Slide Images (WSIs) in which representative tumor regions were annotated by experts and tiled into non-overlapping, $256 \times 256$ pixel patches at a resolution of 0.5 $\mu$m/pixel (downloaded from `https://huggingface.co/datasets/dakomura/tcga-ut`).

Patch-level images were converted into dense numerical representations to capture localized morphology. We used a ResNet-50 convolutional neural network architecture [14], pre-trained on ImageNet [15], as a deterministic frozen feature extractor. The terminal fully connected classification layer was removed, allowing the network to map each $256 \times 256$ pixel histopathology patch into a 2048-dimensional visual feature vector. These raw visual features were extracted across the full cohort and used as inputs to the trainable image encoder.

*b) **Transcriptomic modality processing**:* Bulk RNA-seq gene expression data, normalized via RNA-Seq by Expectation-Maximization (RSEM) [16], were obtained from cBioPortal [17]. To reduce noise and computational complexity, the analysis was restricted to the top 2,000 most variable genes. To prevent data leakage, gene variance calculations were performed exclusively within the training partition. The RSEM expression values were then $\log(x+1)$-transformed to reduce skewness, stabilize variance, and support neural network training.

### *B. Cross-modal alignment framework*

To learn the correspondence between tissue morphology and gene expression, we designed a two-tower dual-encoder architecture that processes histopathology and transcriptomic inputs independently before projecting them into a shared embedding space.

*a) **Image encoder**:* Before entering the trainable neural network, patch-level visual features were aggregated into a patient-level representation. During training, each batch includes all patients in the training split. For each patient, a random subset of $K_{\text{patch}} = 16$ histopathology patches is selected, and their corresponding 2048-dimensional features are aggregated by mean pooling. This stochastic patch sampling increases the variety of pooled visual representations across training steps and acts as a form of data augmentation. For validation and testing, all available patches for each patient are mean-pooled to obtain a deterministic patient-level visual representation.

The aggregated patient-level visual vector is passed to a multi-layer perceptron (MLP) consisting of two hidden fully connected layers with dimensions 1024 and 512. Each hidden layer is followed by batch normalization, ReLU activation, and dropout with probabilities of 0.2 and 0.3, respectively. This encoder maps the input visual features to a 128-dimensional embedding vector $z_v \in \mathbb{R}^{128}$.

*b) **Transcriptomic encoder**:* The input to the transcriptomic tower is the 2,000-dimensional $\log(x+1)$-transformed gene expression vector. We use an MLP with the same architecture as the image encoder, including hidden dimensions of 1024 and 512 with batch normalization, ReLU activation, and dropout. This encoder maps the input gene expression features to a 128-dimensional embedding vector $z_g \in \mathbb{R}^{128}$, matching the dimensionality of the visual embedding space.

### C. Model training and implementation

*a) **Contrastive learning objective**:* To align the two modalities, we use the Normalized Temperature-scaled Cross Entropy loss (InfoNCE) [18]. The objective maximizes the cosine similarity between image and gene embeddings from the same patient while minimizing the similarity between mismatched image–gene pairs within a batch. For a batch of size $B$, the directional losses for the $i$-th patient pair are defined as:

$$\mathrm{L}_i^{v \to g} = -\log \frac{\exp(\mathrm{sim}(z_v^i, z_g^i)/\tau)}{\sum_{j=1}^{B} \exp(\mathrm{sim}(z_v^i, z_g^j)/\tau)} \tag{1}$$

$$\mathrm{L}_i^{g \to v} = -\log \frac{\exp(\mathrm{sim}(z_g^i, z_v^i)/\tau)}{\sum_{j=1}^{B} \exp(\mathrm{sim}(z_g^i, z_v^j)/\tau)} \tag{2}$$

where $\mathrm{sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$ denotes cosine similarity and $\tau$ is the temperature parameter. The final batch-level loss is the average of the two directional losses across all patient pairs in the batch.

*b) **Validation and testing protocol**:* To ensure stable and reproducible evaluation during training, we construct a single comprehensive validation batch at the end of each epoch. Unlike the stochastic patch sampling used during training, validation and testing use global mean pooling across *all* available histology patches for every patient to generate deterministic visual representations.

*c) **Implementation details**:* The framework is implemented in PyTorch and trained using PyTorch Lightning. We use the AdamW optimizer with a CosineAnnealingLR learning rate scheduler and a minimum learning rate of 1e-7. Hyperparameter optimization was performed using Optuna [19]. The tuned hyperparameters were learning rate = 1.88e-5, weight decay = 4e-3, and temperature ($\tau$) = 0.217. Each training step contains all training patients with a newly sampled subset of histopathology patches; one epoch is defined as 20 such stochastic resampling steps. The network is trained for 100 epochs on a single NVIDIA GPU. The complete source code is available on GitHub to ensure reproducibility.[1]

### D. Downstream evaluation tasks

After training, the dual-encoder framework is used to evaluate cross-modal retrieval on held-out test patients. Test-set histopathology features are projected into the shared embedding space using global mean pooling and the trained image encoder. We evaluate the framework across three downstream applications. In the downstream retrieval analyses, $K$ denotes the number of nearest transcriptomic neighbors retrieved from the reference embedding space; this is distinct from $K_{\text{patch}} = 16$, the number of histopathology patches sampled per patient during training.

*a) **Gene expression retrieval**:* Using the visual embeddings of held-out test patients, we compute cosine similarity against the pool of train and validation transcriptomic embeddings, which serve as the reference set. For each test sample, we identify its $K$-nearest transcriptomic neighbors. The retrieved expression value for each gene is estimated as the median expression value of that gene across the $K$ neighbors.

*b) **Gene set enrichment analysis**:* To evaluate pathway-level concordance of the retrieved transcriptomic profiles, we perform single-sample gene set enrichment analysis (ssGSEA) [20]. Enrichment scores are calculated for both the true and retrieved expression profiles across all test samples. For input gene sets, we use the 50 Hallmark gene sets from the Molecular Signatures Database (MSigDB) [21].

*c) **Tumor subtype identification**:* PAM50 intrinsic breast cancer subtype labels obtained via cBioPortal are assigned to patients as ground-truth labels for downstream evaluation [22]. These labels are not used during contrastive representation learning. To classify the tumor subtype of a test patient using only histopathology, its visual embedding is aligned to the reference space to find the $K$-nearest transcriptomic neighbors. The predicted subtype is determined by majority vote among the labels of these neighbors. If the vote results in a tie, the sample is designated as *Unclassified*.

## III. Results and Discussion

**The learned alignment is preserved in held-out test patients.** We first evaluated whether the learned embeddings preserved cross-modal alignment in held-out TCGA-BRCA patients. As shown in Figure 2, the observed validation and test contrastive losses were lower than the distributions obtained from randomly generated embeddings with matched validation and test sample sizes. This indicates non-random alignment between histopathology and transcriptomic representations in held-out patients. Because this evaluation is

[1] `https://github.com/sogandzk/HistoGenX.git`

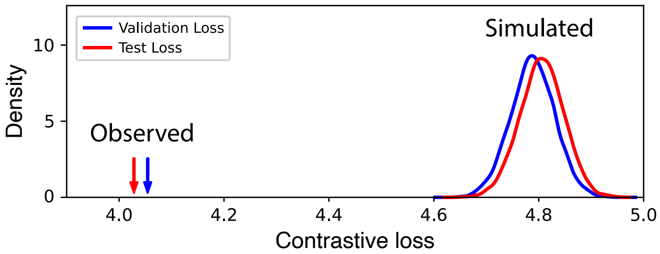


**Figure 2:** Simulated losses were computed from randomly generated embeddings with the same validation and test sample sizes across 10,000 repetitions. Arrows indicate the observed validation and test losses obtained from the trained model's learned histopathology and transcriptomic representations.

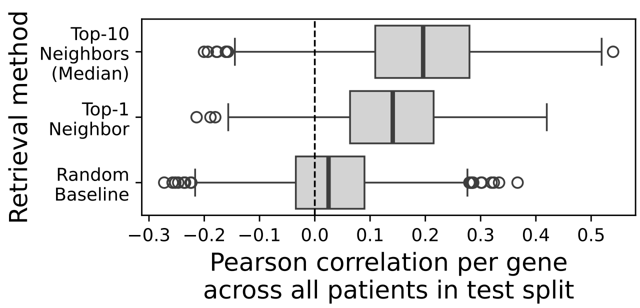


**Figure 3:** Distributions represent the gene-wise Pearson correlations calculated between the retrieved and true expression for the 2,000 most variable genes across the test set patients. Three strategies are compared: (i) Top-10 Neighbors, utilizing the median expression of the ten nearest alignments; (ii) Top-1 Neighbor, utilizing only the single nearest alignment; and (iii) Random Baseline, representing correlations achieved via stochastic alignment.

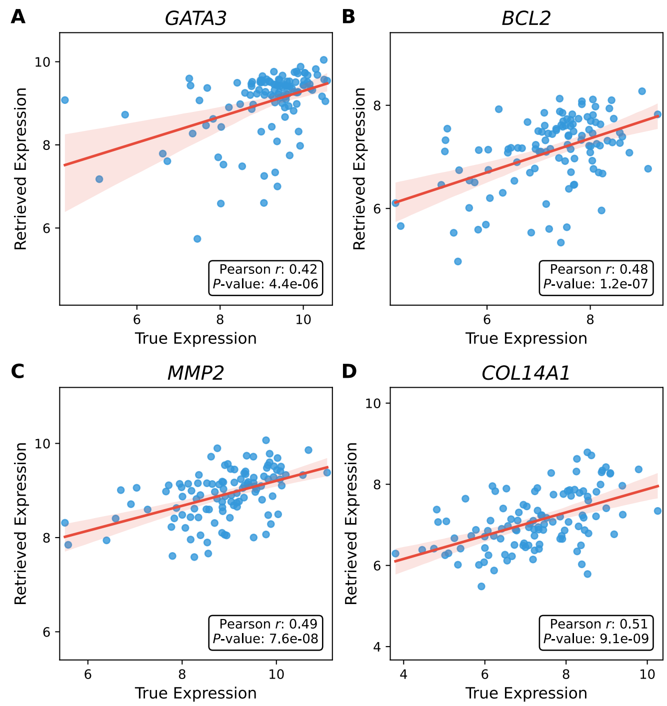


**Figure 4:** Selected biologically relevant genes illustrate examples of positive gene-level concordance between measured and retrieved expression profiles. Scatter plots show ground-truth expression on the x-axis and model-retrieved expression on the y-axis across held-out test patients for: **(A)** *GATA3*, **(B)** *BCL2*, **(C)** *MMP2*, and **(D)** *COL14A1*. Pearson correlation coefficients ($r$) and associated p-values are displayed. The red line represents the linear regression fit with 95% confidence interval.

performed within TCGA-BRCA, this result should be interpreted as internal held-out performance rather than external cohort generalization.

**Gene expression retrieval.** We evaluated the model's ability to retrieve transcriptomic profiles by comparing retrieved gene expression values against the observed ground truth for the 2,000 highly variable genes used during model development. As shown in Figure 3, the retrieved expression profiles showed higher gene-wise concordance with measured expression than the random alignment baseline. Compared with random alignment, which achieved a median Pearson correlation of $r = 0.02$, the retrieved profiles achieved a median Pearson correlation of $r = 0.19$. This demonstrates above-random but modest gene-level concordance between histopathology-derived embeddings and measured transcriptomic profiles.

To provide interpretable examples of gene-level concordance, we visualized retrieved versus measured expression for selected breast-cancer-relevant genes in Figure 4. These examples are intended to illustrate positive associations within the learned embedding space, while the distribution in Figure 3 summarizes performance across all 2,000 genes. The selected genes include *GATA3*, a marker of luminal differentiation in breast cancer, as well as *BCL2*, *MMP2*, and *COL14A1*, which are associated with apoptosis regulation, extracellular matrix remodeling, and stromal or mesenchymal programs, respectively [23–26]. These results suggest that the shared embedding space captures a subset of morphology-associated transcriptomic signals relevant to breast cancer biology, including epithelial lineage identity, apoptosis-related signaling, and stromal interactions. However, because performance varies across genes, the gene-level results should be interpreted as partial transcriptomic retrieval rather than complete transcriptome reconstruction.

**Pathway activity retrieval.** We next investigated whether the retrieved expression profiles preserved pathway-level biological structure. For this analysis, single-sample gene set enrichment analysis (ssGSEA) scores were computed for the 50 Hallmark gene sets from the Molecular Signatures Database (MSigDB) using both true and retrieved expression profiles. As shown in Figure 5, several Hallmark gene sets showed positive concordance between true and retrieved enrichment scores, with a median Pearson correlation of $r = 0.27$ across the 50 pathways. Significant correlations after FDR correction are highlighted in Figure 5.

Although the MSigDB Hallmark collection is distinct from the conceptual Hallmarks of Cancer framework, several Hallmark gene sets capture biological programs relevant to cancer progression [21, 27]. Representative pathways with positive concordance include mTORC1 signaling, early estrogen response, and E2F targets, shown in Figure 5B–D. These pathways are biologically relevant because mTORC1 signaling is associated with cell growth, protein synthesis, early estrogen response reflects estrogen receptor (ER)-driven transcriptional activity characteristic of hormone receptor-positive tumors, and E2F target activation is linked to cell-cycle progression and proliferative capacity. Importantly, these results indicate concordance with transcriptomic pathway scores and should not be interpreted as direct evidence of protein-level pathway activation, therapeutic sensitivity, or clinical actionability. Compared with individual genes, pathway scores summarize coordinated activity across groups of genes and may therefore provide a more stable and interpretable view of the molecular information captured by histopathology-derived embeddings.

**Tumor subtype retrieval.** Finally, we evaluated whether the learned histopathology embeddings contained subtype-relevant molecular information. PAM50 subtype retrieval was performed using the $K$-nearest neighbor retrieval procedure in the shared embedding space. Importantly, PAM50 subtype labels were not used during contrastive representation learning; they were used only for downstream nearest-neighbor subtype assignment.

Figure 6 illustrates the subtype retrieval performance across values of $K$ from 1 to 10. Figure 6A reports overall classification metrics, including macro-, weighted-, and micro-averaged F1-score, precision, and recall. Based on a combined view of the overall metrics, we decided to choose $K = 4$ for presenting the classification. Figure 6B reports subtype-specific F1-scores across values of $K$, and Figure 6C shows the confusion matrix for the representative setting $K = 4$. Although the cohort is predominantly Luminal A, the model identified several Basal and HER2 cases. At the same time, cross-subtype misclassifications remained, indicating that subtype retrieval is imperfect. These results suggest that the learned embedding space retains subtype-relevant molecular structure, but the PAM50 analysis should be interpreted as an exploratory downstream evaluation rather than a validated clinical subtype classifier.

Taken together, these results support the feasibility of patient-level contrastive alignment between breast cancer histopathology features and matched bulk RNA-seq profiles. The model retrieves approximate gene-expression profiles, preserves selected pathway-level signals, and supports molecular subtype retrieval from histopathology-derived embeddings. The results do not indicate that the current model can replace molecular testing or clinical pathology workflows. Rather, they provide a reproducible proof of concept for image-based transcriptomic profiling and molecularly informed retrieval from routine H&E histopathology.

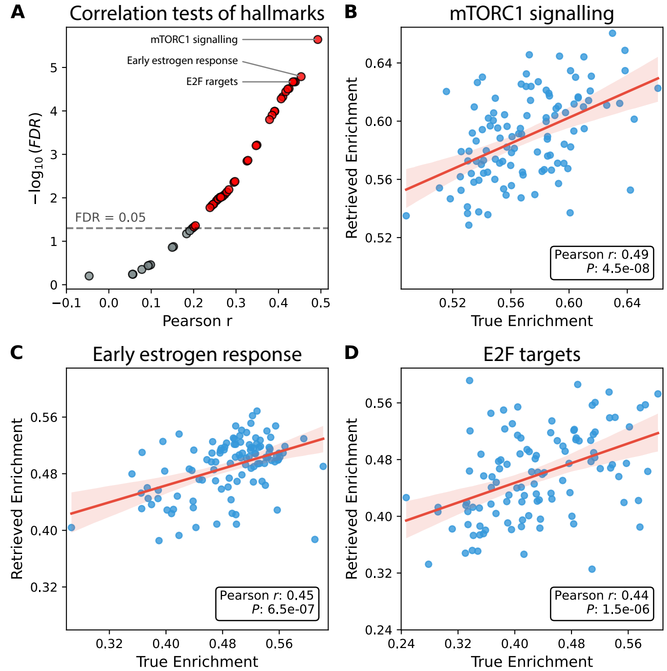


**Figure 5:** Correlation analysis of ssGSEA enrichment scores of the 50 hallmark gene sets from MSigDB calculated based on true versus retrieved gene expression profiles across test set patients. **(A)** Summary of the correlation tests across all hallmark gene sets. The y-axis represents -log10(FDR) and the x-axis represents Pearson $r$ values. Significant correlations (FDR $< 0.05$) are highlighted in red, with the three representative gene sets shown in **(B-D)** annotated. **(B-D)** Scatter plots of enrichment scores based on retrieved (y-axis) versus true (x-axis) gene expression profiles for: **(B)** mTORC1 signaling, **(C)** Early estrogen response, and **(D)** E2F targets. Pearson correlation coefficients ($r$) and associated p-values are displayed. The red line represents the linear regression fit with 95% confidence interval.

## IV. Limitations and Future Work

Several limitations should be considered. The current evaluation is based on held-out TCGA-BRCA patients and does not include a fully independent external cohort. The workflow also relies on pre-annotated TCGA-UT tumor-region patches. Applying the framework to external cohorts with raw whole-slide images would require reliable tumor-region selection or segmentation before feature extraction.

The transcriptomic target is bulk RNA-seq, which provides patient-level molecular supervision but cannot localize gene expression to individual patches or spatial tissue regions. Because bulk RNA-seq averages signals across tumor, stromal, immune, and other cell populations, some learned image–transcriptome associations may reflect tumor microenvironment composition rather than tumor-intrinsic biology alone. In addition, RNA expression does not necessarily correspond to protein abundance or functional pathway activation. Future work should evaluate protein-level markers, proteomic data, or multiplex imaging where clinically relevant.

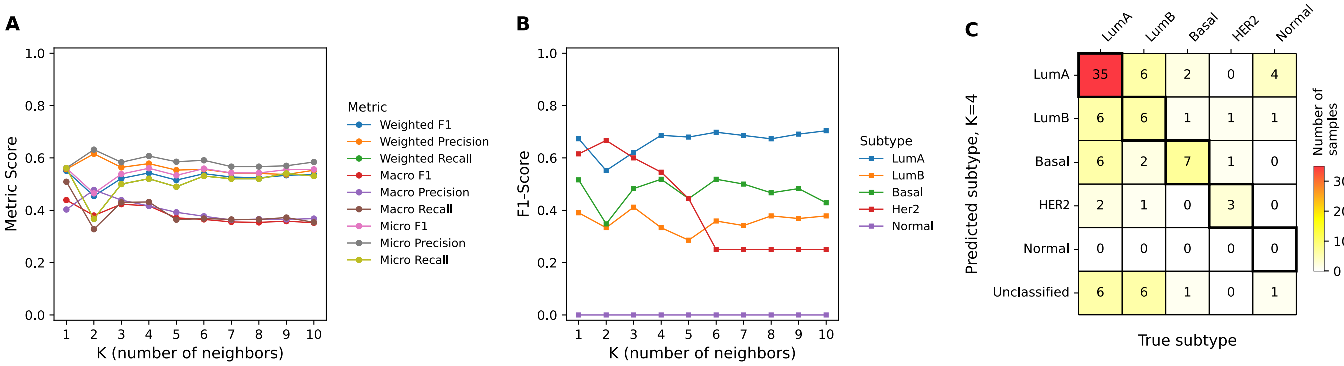


**Figure 6:** PAM50 subtype retrieval performance. **(A)** Overall classification performance metrics, including F1 score, precision, and recall computed using weighted, macro, and micro averaging schemes, evaluated across different values of $K$ used in the $K$-nearest neighbor retrieval method. **(B)** Subtype-specific classification F1-scores across different values of $K$. **(C)** Confusion matrix of the classification results for $K = 4$, illustrating the distribution of predicted versus true subtype labels. Here, $K$ denotes the number of nearest transcriptomic neighbors retrieved from the reference embedding space.

Histopathology images are affected by pre-analytical and technical variation, including fixation, staining, scanner, and site-specific differences. External validation and robustness analyses across institutions and slide-preparation workflows are therefore necessary before clinical translation. A related limitation is the potential spatial mismatch between the diagnostic slide and the tissue aliquot used for bulk RNA-seq, particularly in heterogeneous tumors with regional variation in grade, stromal density, necrosis, or molecular subclones.

The image representation relies on fixed ImageNet-pretrained ResNet-50 features and mean pooling, which may discard spatial organization and subtle pathology-specific morphology. PAM50 subtype retrieval is also affected by class imbalance and should be interpreted as an exploratory downstream evaluation rather than a validated clinical classifier. Future work should evaluate independent cohorts, pathology-specific foundation models, attention-based whole-slide aggregation, spatial transcriptomic supervision, calibration, and clinically relevant endpoints.

## V. Conclusion

We presented a patient-level dual-encoder contrastive learning framework for aligning breast cancer histopathology features with matched bulk RNA-seq transcriptomic profiles. The learned embedding space supports cross-modal retrieval and shows concordance with gene-expression, MSigDB Hallmark pathway, and PAM50 subtype information from histopathology-derived features. These results provide a reproducible proof of concept for image-based transcriptomic profiling from routine H&E histopathology. Future work should focus on external validation, pathology-specific visual encoders, attention-based whole-slide aggregation, and spatially resolved molecular supervision.

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## Ethics Statement

This study used publicly available, de-identified data from TCGA-BRCA, TCGA-UT, and cBioPortal. No new human-subject data were collected by the authors. The original data-generating studies obtained ethical approval and informed consent according to their respective institutional requirements. This secondary analysis of de-identified public data did not involve additional human-subject procedures.

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